RAJALAKSHMI ENGINEERING COLLEGE

Rakchena C 250701576-2025-CSE

R2

Started on	Wednesday, 23 September 2026, 3:29 PM
State	Finished
Completed on	Wednesday, 23 September 2026, 3:36 PM
Time taken	7 mins 21 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2		**g**	x	**t**	**x**	**a**	y	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<string.h>
int max(int a,int b){
    return (a>b)?a:b;
}
int main(){
    char s1[100],s2[100];
    int dp[101][101];
    int n,m,i,j;
    scanf("%s",s1);
    scanf("%s",s2);
    n=strlen(s1);
    m=strlen(s2);
    for(i=0;i<=n;i++){
        dp[i][0]=0;
    }
    for(j=0;j<=m;j++){
        dp[0][j]=0;
    }
    for(i=1;i<=n;i++){
        for(j=1;j<=m;j++){
            if(s1[i-1]==s2[j-1]){
                dp[i][j]=dp[i-1][j-1]+1;
            }else{
                dp[i][j]=max(dp[i-1][j],dp[i][j-1]);
            }
        }
    }
    printf("%d",dp[n][m]);
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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